

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 09)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

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0 1 . 2

Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

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0 1 . 3

Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

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0 1 . 4

Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

10

0 2 . 1

Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

0 2 . 2

Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

1

2

3

0 2 . 3

Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores

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Gas analysis of ice cores

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Radiocarbon dating

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Magnesium : calcium ratio in calcite

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10

0 3 . 1

Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station

☐

B Flooding a valley to create a reservoir

☐

C Lowering the water table by groundwater abstraction

☐

D Acid mine drainage entering a stream

☐

0 3 . 2

Give **four** human activities that cause complete habitat destruction.

[4 marks]

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5

Discuss the strategies that can be used to increase the sustainability of agriculture.

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Extra space

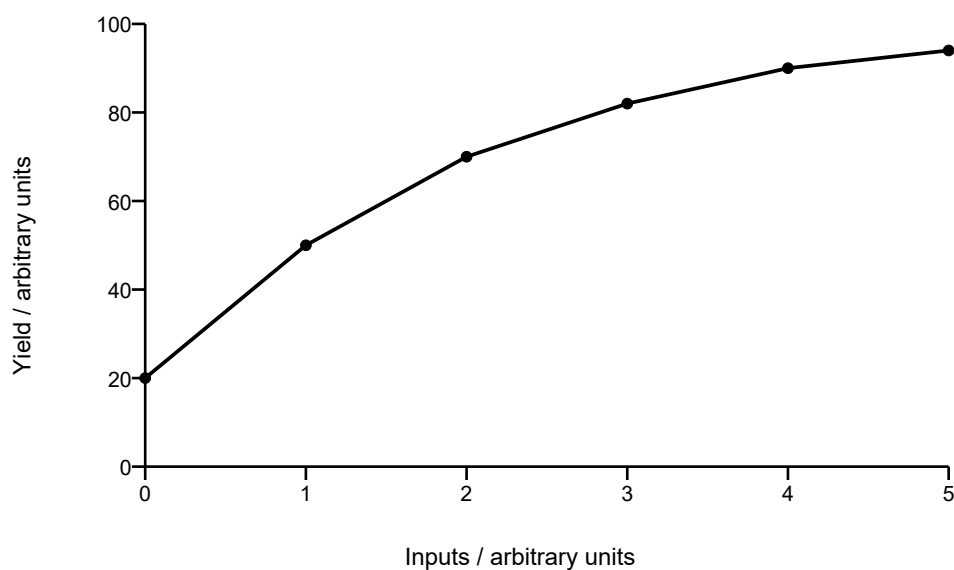
04 . 2 State what is meant by **permaculture**.

[1 mark]

10

0 5

Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1**0 5****. 1**

Describe the relationship shown in Figure 1.

[2 marks]**0 5****. 2**

Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]**Answer**

units

0 5**. 3**

Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

0 5 . 4

Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

0 5 . 5

Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

10

0 6

A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 6 . 1

Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size

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Distribution

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Survival rate

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Age structure

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0 6 . 2 Describe how the population size of the antelope could be estimated.
[2 marks]

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10

07 . 1

Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

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07 . 2

Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

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07 . 3

Explain **three** ways in which coral reefs are important to humans.

[3 marks]

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08

Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha ⁻¹	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

08

1

Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

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08

2

Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

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08 . 3

Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

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09

Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

09 . 1

Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

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09 . 2

Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

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09 . 3

Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

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09 . 4

Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

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10

10 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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[illegible]

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10 . **2** State what is meant by a **fed** aquaculture species. **[1 mark]**

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS